Matteo Tomasini, PhD

Researcher in digital and computational archaeology

Institute for Archaeological Sciences
University of Bern, Switzerland

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21 November 1989
Born in Locarno, TI
Swiss citizenship

PROFESSIONAL EXPERIENCE

Senior assistant — Bern, Switzerland
Universität Bern — 2026 – current

- Coordination of Computational and Digital Archaeology Lab
- Research in computational archaeology

Research engineer — Gothenburg, Sweden
Göteborgs Universitet — 2022 – 2026

- Interdisciplinary support to researchers in archaeology and the humanities
- Project management and development tasks in scientific visualization as a member of the Swedish Infrastructure for Scientific Visualization (InfraVis)

Postdoctoral research fellow — Gothenburg, Sweden
Göteborgs Universitet — 2021 – 2022

- Development of mathematical model in marine evolutionary biology

Postdoctoral research associate — East Lansing, USA
Michigan State University — 2019 – 2020

- Development of statistical algorithms in population genetics

SKILLS

Interdisciplinary research · Project management · Teaching · Networking · Effective communication · Scientific writing · Leadership

Computational archaeology · Research software development · Mathematical modelling · Applied statistics · Data pipelines · Data management · GIS · Spatial statistics · Evolutionary biology · Physics

Italian · mother tongue
French · proficient
English · proficient
German · intermediate
Swedish · intermediate
Spanish · basic

CMAS One Star Diver
CMAS Nitrox Diver

EDUCATION

PhD in ecology and evolution — Bern, Switzerland
Universität Bern — 2015 – 2019

- Thesis: *On the theory of rapid adaptation in the presence of gene flow*

BSc/MSc in theoretical physics — Geneva, Switzerland
Université de Genève — 2009– 2014

Swiss matura, Latin & Greek — Locarno, Switzerland
Liceo di Locarno — 2004 – 2008

APPOINTMENTS

Convener	CAA special interest group in scripting languages in archaeology
Board Member	Nordic Research Software Engineers
Advocate	Dark Sky International

TEACHING EXPERIENCE (SELECTION)

Vibe Coding May 2026
Session/workshop on coding with generative AI at GRIDH's *Tech Thursday* events. Workshop

Code review and best practices in programming Mar 2026
Workshop at CAA Conference 2026, Vienna, Austria. Workshop

Introduction to agent-based modelling Nov 2025 / May 2026
Part of course "Digital Methods and Cultural Heritage" at University of Gothenburg. Lesson

Bases of scientific visualization for political scientists Nov 2025
Invited by Department of Political Sciences at University of Gothenburg.
Co-hosted with Dr. Westin and Dr. Alfter. Course

Basics of visualization in R Jun 2025
Internal course at InfraVis. Course

Applied Biostatistics I & II Sep 2017 – Jun 2019
Teaching assistant and substitute teacher for MSc course at University of Bern. Course

FIELD WORK

University Aula, Oslo, Norway Jun 2026
Digitization of Edvard Munch's large mural paintings of the University Aula at the University of Oslo

Per Dubbshuset, Gothenburg, Sweden Apr 2024
Digitization for preservation of the Per Dubbshuset wing at Sahlgrenska Hospital

Le Yaudet / Landéda, Bretagne, France Aug / Sep 2023
Tests of navigation performance for prehistoric boats reconstructions

CONFERENCE AND EVENTS ORGANIZATION

Journal club in Archaeology, Genetics, and Culture 2025 – active
Organizer and host

Nordic-RSE 2025 Conference May 2025
University of Gothenburg, Sweden. Local organizing chair.

Nordic-RSE 2024 Conference May 2024
Aalto University, Espoo, Finland. Chair of organizing committee.

CeMEB General Assembly Oct 2022
Tjärnö Marine Laboratory, University of Gothenburg, Sweden. Member of organizing committee.

IT STACK

Software development Python, (C++, Matlab)
Development of custom-made software for projects of climatology, evolutionary biology, digital humanities, and computational archaeology

Data management systems Django, PostGIS, DBeaver, pgAdmin
Development and maintenance of back-ends serving georeferenced databases composed of multi-media data (including 3D models, images, text, metadata)

Data analysis and statistics Python (Pandas, Seaborn), R, Stan

Development of data analysis pipelines, from performance of exploratory data analysis, to publication-level results, including Baeysian statistics

Continuous integration / Continuous Delivery processes Git, Github, Pytest, Sphinx, Conda
Usage of standard tooling for CI/CD practices to ensure prompt delivery of updates on live computational systems

FURTHER EDUCATION

Eurodoc ValDem Ambassador Programme Spring 2026
Programme about research, higher education governance, and democratic processes, organised by *The European Council of Doctoral Candidates and Junior Researchers (Eurodoc)*

ML / AI Spring 2018
Practical course in machine learning and deep neural networks (given by Dr. Haug at University of Bern)

Machine Learning Autumn 2017
Master level course in machine learning (given by Prof. Wegmann at University of Fribourg)

MILITARY SERVICE

500 days of service between 2008 and 2019. Released from service at the end of 2025 for age limits.

Chief sergeant major Bière, Simplon, Frauenfeld
Gruppo Artiglieria 49, Batterie Art 1 e 3 (WK / CR) 2011 – 2019

NCO school Sion
Stage de formation de sous-officiers superieurs 2009

Boot camp Bière
École de recrues d'artillerie 31 2008 – 2009

ACADEMIC PRODUCTION

Peer-reviewed papers

1. J. Rieder, M. Jahnke, …, **M. Tomasini**, F. A. M. Volckaert (2025), *Seascape genomics: assisting marine biodiversity management by combining genetic knowledge with environmental and ecological information*, Marine Policy, Volume 182. 10.1016/j.marpol.2025.106867

2. B. Bengtsson, A. Montenegro, A. Green, **M. Tomasini**, M. Prince, V. Wåhlstrand Skärström, K. I. Austvoll, J. Ling, C. Lindhé (2025), *Seafaring and navigation in the Nordic Bronze Age: The application of an ocean voyage tool and boat performance data for comparing direct open water crossings with sheltered coastal routes.* PLOS One. 10.1371/journal.pone.0320791

3. **M. Tomasini**, S. Peischl (2022), *The role of spatial structure in multi-deme models of evolutionary rescue*, Journal of Evolutionary Biology, 35(7), 986 – 1001. 10.1111/jeb.14018

4. **M. Tomasini**, S. Peischl (2020), *When does gene flow facilitate evolutionary rescue?*, Evolution, 74(8), 1640 – 1653. 10.1111/evo.14038

5. **M. Tomasini**, S. Peischl (2018), *Establishment of locally adapted mutations under divergent selection*, Genetics, 209(3), 885 – 895. 10.1534/genetics.118.301104

Book chapters and conference proceedings

1. B. Bengtsson, M. Prince, A. Montenegro, **M. Tomasini** (accepted), *Testing the digital reconstruction of the Varpelev Boat using predicted boat performance data and agent-based simulations*, in M. Ravn and O. T. Kastholm, *The Varpelev Boat*, The Vikings Ship Museum, Roskilde.

2. B. Bengtsson, J. Ling, A. Montenegro, **M. Tomasini** (accepted), *Exploring the western maritime trade route: how Great Orme copper (and Cornish tin) made it into Scandinavia c.1700–1400 BC*, in J. Koch, M. Fauvelle, B. Cunliffe, and J. Ling, *Moving Metals by Sea: Great Orme and the Bronze Age World*, Oxbow Books

3. J. Westin, C. Lindhé, D. Brodén, G. Almevik, **M. Tomasini**, (2025) *DigiCURE: Building a Digital Humanities Infrastructure for Preserving and Studying At-risk Cultural Heritage*, Proceedings of the second HumInfra Conference (HiC 2025), 115 – 121.

4. J. Westin, T. Bridge, **M. Tomasini** (2024), *From the Arctics to Antarctica – A multimodular visualisation of data.* Proceedings of the Huminfra Conference (HiC 2024), 135 – 140. 10.3384/ecp205018

Under review

1. M. Vander Linden, **M. Tomasini** (under review) *Simulating a (Bell Beaker) metapopulation: theoretical and methodological implications for analysing the archaeological record*, Proceedings of the 23rd Archéologie & Gobelets Conference, Vannes, 2026

Software and web applications

1. J. Westin, T. Bridge, **M. Tomasini** (2025), *Etruscan Chamber Tombs* (Version 1.6) [Web portal]. https://etruscan.dh.gu.se/.

2. **M. Tomasini** (2025), *Pytheas Simulator* (Version 1.0) [Python library]. https://github.com/mtomasini/pytheas-simulator.

3. **M. Tomasini** (2024), *MetaPypulation* (Version 0.1) [Python library]. https://github.com/mtomasini/MetapopulationsPython.

4. J. Westin, T. Bridge, **M. Tomasini**, A. Green, A. Karimi (2024), *Saint Sophia's Inscriptions* (Version 1.6) [Web portal]. https://saintsophia.dh.gu.se/.

5. J. Westin, T. Bridge, **M. Tomasini** (2024), *Per Dubbshuset portal* (Version 1.0) [Web portal]. https://perdubbshuset.dh.gu.se/.

Oral contributions (selection)

1. Session organizer: **"All Models Are Wrong": Learning from Failure in Computational Archaeology**, *CAA International conference 2026*, Vienna, Greece (with J. Roe)

2. **From navigation to polar diagrams: a case study**, *CAA International conference 2026*, Vienna, Austria (with B. Bengtsson, B. Bengtsson, A. Burlot, P. Guillonnet, A. Montenegro, M. Prince, F. Vény)

3. **Pytheas: a fast and agile ABM software for the study of ancient seafaring**, *CAA International conference 2025*, Athens, Greece (with A. Montenegro, B. Bengtsson)

4. **What can metapopulation models say about maritime connectivity in ancient times?**, Kiel Conference 2025, Kiel, Germany (with M. Vander Linden)

5. Invited seminar: **Computer simulated ocean voyages between Scandinavia and Great Britain during the Bronze Age**, at Department of Historical Sciences, University of Gothenburg, Sweden (with A. Montenegro)

6. Session organizer: **Evolution.**, Theoretical Archaeology Group 2024, Bournemouth, United Kingdom (with M. Vander Linden, E. Crema)

7. Featured session organizer: **Fight or flight? The role of adaptation in shaping future species' distributions in light of climate change**, YOUMARES 13, Berln, Germany (with M. Eriksson)

8. **How do species ranges respond to the effects of counteracting environmental gradients?**, European Society for Evolutionary Biology Congress 2022, Prague, Czech Republic (with M. Rafajlović, M. Eriksson, K. Johannesson)

9. **Range expansions along multiple environmental gradients**, Baltic Sea Science Congress 2021, Aarhus, Denmark (with M. Rafajlović)

10. **Effects of gene flow and fragmentation on evolutionary rescue** Modelling Ecology & Evolution Zurich 2019, Zurich, Switzerland

Funding

Bando Regione Toscana "Seal of Excellence – MSCA-PF 2025" 134,500€
Bundles of variation: interpreting transmission of material culture in the Neolithic of the Central and Western Mediterranean (Refused)
Kungl. Vetenskaps- och Vitterhets-Samhället i Göteborg (KVVS) Stipendium 2022 9,500 SEK

Intl. Council for the Exploration of the Sea (ICES) Early Career Scientist 2021 750€

Other awards

Marie Skłodowska-Curie Actions – Individual Fellowship 2025 Seal of Excellence
Bundles of variation: interpreting transmission of material culture in the Neolithic of the Central and Western Mediterranean, Evaluation Score: 91.00%

Marie Skłodowska-Curie Actions – Individual Fellowship 2020 Seal of Excellence
The evolution of marine species in continuous space, Evaluation Score: 90.40%

Scientific committees and appointments

Ambassador 2026 – current
Eurodoc Ambassador Programme for Values and Democracy in Research and Higher Education

Convener 2026 – current
Scientific Scripting Languages in Archaeology, Special Interest Group of the *Computer Applications and Quantitative Methods in Archaeology* association

Advocate 2025 – current
Dark Sky International; local events organiser for Dark Sky Sweden

National representative (Sweden) 2025 – current
International Council of RSE Associations

Ambassador 2024 – 2026
CodeRefinery

Board member 2022 – current
Nordic-RSE; chair between 2023 and March 2026

Steering committee 2021 – 2023
Linnaeus Centre for Marine Evolutionary Biology